

Problem-Solution Fit

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit :

1. CUSTOMER SEGMENT(S) [CS] • Farmers (small,medium, and large-scale) • Agricultural Researchers • Agricultural Extension Officers • Policymakers and Government Agriculture Departments	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited technical knowledge • Limited access to agricultural experts • Budget constraints • Limited internet connectivity in rural areas	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional farming knowledge: Easy to follow but subjective and less accurate. Agricultural experts: Reliable but expensive and not always available. Online articles: Easily accessible but generic and not personalized.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty selecting the most suitable crop (Very High) • Low crop yield due to poor decisions (High) • Inefficient use of fertilizers and water (High) • Lack of data-driven agricultural guidance (High)	9. PROBLEM ROOT / CAUSE [RC]	7. BEHAVIOR + ITS INTENSITY [BE] • Relies on experience and traditional practices (High) • Consults neighboring farmers (Medium) • Searches online for farming advice (Medium) • Willing to adopt simple technology that provides reliable recommendations (High)
3. TRIGGERS TO ACT [TR] • Beginning of a new cultivation season • Poor harvest in previous seasons	10. YOUR SOLUTION [SL] OptiCrop is a machine learning-based smart agricultural recommendation system that analyzes soil	8. CHANNELS OF BEHAVIOR [CH] ONLINE Web application, Agricultural websites, Government agriculture portals, Mobile internet

• Changes in soil fertility or weather conditions • Need to maximize productivity and profits	nutrients (N, P, K) and environmental factors (temperature, humidity, pH, rainfall) to recommend the most suitable crop. It provides fast, accurate, and data-driven recommendations through an easy-to-use web application, helping farmers improve crop yield, optimize resource utilization, and support sustainable agriculture.	
4. EMOTIONS BEFORE / AFTER [EM] Before: Confused, Uncertain, Worried, Frustrated After: Confident, Informed, Satisfied, Optimistic		OFFLINE Local agriculture offices, Farmer meetings, Extension officers, Community farming groups